COMSATS University Islamabad

Attock Campus

Department Of Computer Science

Course	**IS (LAB)**
Instructor	**Ms. Ambreen Gul**
Lab Tssks	

Student Details

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1:

```python
def gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return a
print(gcd(48, 18))
```

Output:

```
PS D:\Mids> python -u "d:\Mids\Assignment.py"
6
PS D:\Mids>
```

2:

```python
def gcd_steps(a, b):
    while b != 0:
        print(f"{a} = {b} * ({a // b}) + {a % b}")
        a, b = b, a % b
    print(a)
    return a

gcd_steps(48, 18)
```

Output:

```
PS D:\Mids> python -u "d:\Mids\Assignment.py"
48 = 18 * (2) + 12
18 = 12 * (1) + 6
12 = 6 * (2) + 0
6
PS D:\Mids>
```

3:

```python
def gcd_recursive(a, b):
    if b == 0:
        return a
    return gcd_recursive(b, a % b)

print(gcd_recursive(48, 18))
```

Output:

```
PS D:\Mids> python -u "d:\Mids\Assignment.py"
6
PS D:\Mids>
```